

Section 10: Dimension Reduction

(UC Davis, Spring 2026) — STA 141A:
Fundamentals of Statistical Data Science

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Overview

Based on Chapter 12 of ISL book James et al. (2021). For more R code examples, see R Markdown files in [book website](#).

Principal component analysis (PCA)

Dimension reduction

Suppose we have n observations on a set of p features $X_1, X_2, \dots, X_p$.

- That is, suppose we have n data points

$$\mathbf{x}_1 = (x_{11}, x_{12}, \dots, x_{1p})^\top, \quad \mathbf{x}_2 = (x_{21}, x_{22}, \dots, x_{2p})^\top, \quad \dots, \quad \mathbf{x}_n = (x_{n1}, x_{n2}, \dots, x_{np})^\top.$$

- *Dimension reduction*: reduce dimensionality of data while retaining as much information about the data as possible.
- Idea: not all p dimensions are equally interesting. If the j th feature has almost the same value for all n observations, do we really need to keep track of the j th feature?
- Dimension reduction is useful for e.g.,
 - 2-dimensional scatterplots of data if $p = 10$.
 - Image compression (reduce computer memory storage requirements).
 - A variational autoencoder (VAE) uses Encoder-Decoder architecture
 - * The Encoder portion compresses high-dimensional data (e.g., thousands of features) into a low-dimensional “latent space.”
 - * The Decoder portion takes a sampled latent representation and attempts to reconstruct the original high-dimensional data from it.

Scaling the variables

We are usually interested in analyzing the impact of certain features in relation to their variation.

- The variance of features can be large solely because their values are large.
- Not an issue if features are measured in the same units.
- So, by scaling (by their standard deviation), the variation among all predictors is comparable, independently of their magnitude.
- In general, scaling the variables to have standard deviation one is recommended.

Some general linear algebra ideas: orthogonal

Consider two vectors $\mathbf{a} = (a_1, \dots, a_p)$ and $\mathbf{b} = (b_1, \dots, b_p)$.

- The *dot product*

$$\mathbf{a} \cdot \mathbf{b} := \sum_{j=1}^p a_j b_j$$

measures how much one vector “points” in the same direction as another.

- The vectors $\mathbf{a}$ and $\mathbf{b}$ are called *orthogonal* or *perpendicular* to each other if and only if the dot product $\mathbf{a} \cdot \mathbf{b}$ is zero.

Some general linear algebra ideas: linear combination

A *linear combination* of the original p features $X_1, X_2, \dots, X_p$ is defined as

$$\phi_1 X_1 + \phi_2 X_2 + \dots + \phi_p X_p$$

for some *coefficient* values $\phi_1, \phi_2, \dots, \phi_p$.

- Such a linear combination can be represented by the vector $\boldsymbol{\phi} = (\phi_1, \phi_2, \dots, \phi_p)$.
- A linear combination is called *normalized* if $\|\boldsymbol{\phi}\|_2^2 = 1$. (unit vector)

Principal component analysis (PCA)

PCA – Idea

PCA seeks a small number of dimensions that are as “interesting” as possible.

- "Interesting" is measured by how much the observations vary along the dimension.
- Each dimension (i.e., *principal component (PC)*) found by PCA is a *linear combination* of the original p features.
- Coefficients $\phi_1, \phi_2, \dots, \phi_p$ of a principal component are called its *loadings*; the vector $\boldsymbol{\phi}$ is called the principal component's *loading vector*.
- Principal component loading vectors are always orthogonal to each other.
- PCA computes p orthogonal principal component loading vectors in order from "most interesting" to "least interesting".

First principal component: how to compute?

Given an $n \times p$ data matrix $\mathbf{X}$, how to compute the *first principal component*?

- Because we are only interested in variance, assume each column in $\mathbf{X}$ has mean zero.
- Coefficients $\phi_{11}, \phi_{21}, \dots, \phi_{p1}$ lead to the n values

$$z_{i1} = \phi_{11}x_{i1} + \phi_{21}x_{i2} + \dots + \phi_{p1}x_{ip} = \sum_{j=1}^p \phi_{j1}x_{ij} \quad i = 1, \dots, n. \quad (1)$$

- The first PC loading vector is defined to be the *normalized* vector $(\phi_{11}, \phi_{21}, \dots, \phi_{p1})$ that maximizes

$$\frac{1}{n} \sum_{i=1}^n z_{i1}^2. \quad (2)$$

- (Normalization ensures the variance is not arbitrarily large.)
- Because each column of $\mathbf{X}$ is centered, mean of $z_{11}, z_{21}, \dots, z_{n1}$ is also zero. So Eq. (2) is (almost) just the sample variance of $z_{11}, z_{21}, \dots, z_{n1}$.
- (2) can be maximized via *eigen decomposition* or *singular value decomposition* (see lab, but don't need to know this for exam).

First principal component: geometric interpretation

The 1st PC loading vector defines a direction in feature space along which the data vary the most.

- For all $i = 1, \dots, n$, the data point $\mathbf{x}_i$ projected onto this direction gives the PC score z_{i1} .

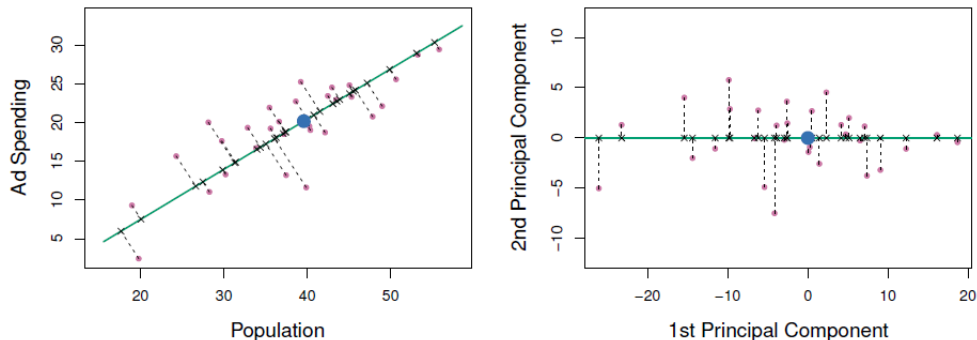


Figure: Figure by James et al. (2021). A subset of the advertising data. The mean pop and ad budgets are indicated with a blue circle. Left: The 1st PC direction (in green). It is the dimension along which the data vary the most, and it defines the line that is closest to all n observations. Right: The left-hand panel has been rotated so that the 1st PC direction coincides with the x-axis.

Remaining principal components

Suppose we have computed the first PC loading vector ϕ_1 . *How to compute the 2nd PC vector?*

- PCA computes the second PC by finding the normalized vector ϕ_2 that maximizes

$$\frac{1}{n} \sum_{i=1}^n z_{i2}^2, \quad (3)$$

but now this vector ϕ_2 must be orthogonal to the vector ϕ_1 .

- PCA maximizes (2) with the additional constraint that ϕ_2 must be orthogonal to ϕ_1 .
- If $p = 2$, then there is only one direction orthogonal to ϕ_1 , and we immediately obtain ϕ_2 .
- If $p > 2$, there are infinitely many directions orthogonal to ϕ_1 .

How to compute the remaining PC vectors? Third, fourth, and so on.

- Once ϕ_2 is computed, PCA finds ϕ_3 by maximizing (2) with the additional constraint that ϕ_3 has to be orthogonal to both ϕ_2 and ϕ_1 .
- And so on to find ϕ_4 , ϕ_5 , etc.

Geometric interpretation: alternative interpretation

PCs provide low-dimensional linear surfaces that are closest to the n observations.

- The first PC loading vector is the line in p -dimensional space closest to the n observations.
- The first two PC loading vectors span the plane closest to the n observations.
- The first three PC loading vectors span the 3-dim hyperplane closest to the n observations.

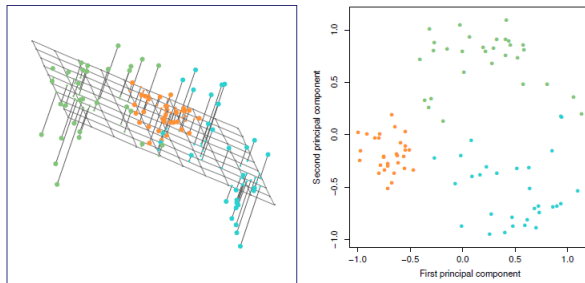


Figure: Figure by James et al. (2021). Observations simulated in three dimensions, displayed in color for ease of visualization. Left: The first two principal component directions span the plane that best fits the data (in sense of minimizing RSS). Right: The first two principal component score vectors give the coordinates of the projection of the observations onto the plane.

Geometric interpretation: alternative interpretation

For any positive integer $M \leq \min\{n - 1, p\}$, together the first M PC score vectors and the first M PC loading vectors provide an M -dimensional approximation to the i th observation:

$$x_{ij} \approx \sum_{m=1}^M z_{im} \phi_{jm}. \quad (4)$$

How good is this approximation? It is the best, in the sense of the following optimization problem.

- Suppose we have a data matrix $\mathbf{X}$ that is column-centered, and have chosen a value of M .
- Of all possible approximations of the form

$$x_{ij} \approx \sum_{m=1}^M a_{im} b_{jm},$$

consider which values $\{a_{im}\}$ and $\{b_{jm}\}$ minimize the sum of squares

$$\sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M a_{im} b_{jm} \right)^2. \quad (5)$$

- It can be shown that for any value of M , the minimizers of (5) are $a_{im} = z_{im}$ and $b_{jm} = \phi_{jm}$.
- When $M = \min\{n - 1, p\}$, the representation (4) becomes exact.

The proportion of variance explained (PVE)

How much information is lost by projecting onto a few PCs? That is, how much variance in the data is not contained in PCs? Here, assume centered data.

- The *total variance* in the data can be decomposed into the sum of two terms:

$$\underbrace{\sum_{j=1}^p \frac{1}{n} \sum_{i=1}^n x_{ij}^2}_{\text{Variance of data}} = \underbrace{\sum_{m=1}^M \frac{1}{n} \sum_{i=1}^n z_{im}^2}_{\text{Var. of first } M \text{ PCs}} + \underbrace{\frac{1}{n} \sum_{j=1}^p \sum_{i=1}^n \left(x_{ij} - \sum_{m=1}^M z_{im} \phi_{jm} \right)^2}_{\text{MSE of } M\text{-dimensional approximation}} \quad (6)$$

- The *proportion of variance explained (PVE)* of the m th PC is the quotient

$$PVE_m = \frac{\text{variance explained by the } m\text{th PC}}{\text{variance of data}} = \frac{\sum_{i=1}^n z_{im}^2}{\sum_{j=1}^p \sum_{i=1}^n x_{ij}^2}. \quad (7)$$

- The proportion of variance explained by the first M PCs is (rearrange terms in (6))

$$\sum_{m=1}^M PVE_m = \frac{\text{variance explained by the first } M \text{ PCs}}{\text{variance of data}} = 1 - \frac{RSS}{TSS}, \quad (8)$$

where TSS is total sum of squared elements of $\mathbf{X}$, and RSS is residual sum of squares of the M -dimensional PC approximation.

PVE is usually visualized with the *screepLOT*

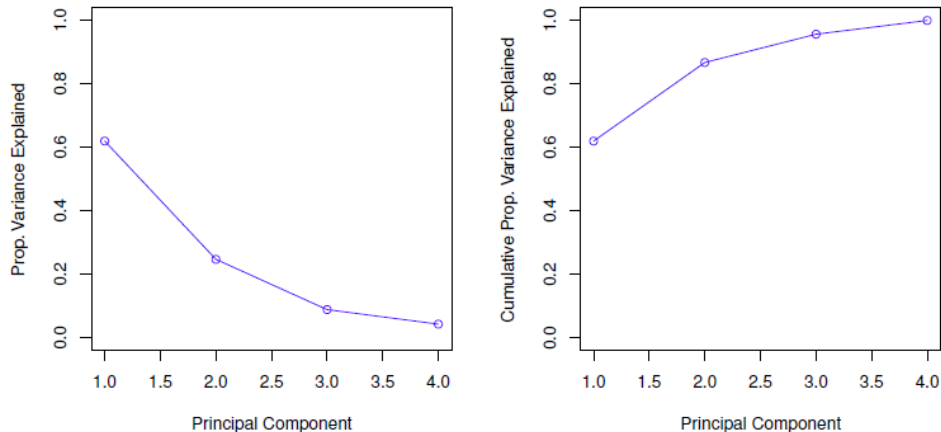


Figure: Figure by James et al. (2021). Left (7): A scree plot depicting the proportion of variance explained (PVE) by each of four principal components in USArrests data. Right (8): The cumulative PVE by four principal components.

Number of predictors

Usually, we only want a few dimensions in order to better visualize or understand the data.

- How many principal components do we need? How can we justify using only e.g., three PCs vs four PCs vs five PCs?
 - There is no simple answer to this! There is no formula that can be applied universally that gives us the optimum value.
 - We can intuitively decide to choose the number of predictors by eyeballing the *screeplot* which depicts the proportion of variance explained (PVE).
 - In the screeplot, we look for a point at which the PVE by each subsequent principal component drops significantly off (this is subjective). Such a drop is often referred to as an *elbow* in the screeplot, and the rule by choosing the point is thus called *elbow rule*.

Other uses for PCA

Many statistical techniques, such as regression, classification, and clustering, use the full $n \times p$ data matrix.

- Can instead use the $n \times M$ matrix whose columns are the first $M \ll p$ PC score vectors.
- This can lead to less noisy results, since often the signal in a data set is concentrated in its first few principal components.